AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM311
Information Security

Final Examination
Summer Semester 2025/2026

Time Allowed: 3 Hours

Instructions:

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **NOT** allowed.

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PART 1: Multiple Choice Questions [20 Marks]

Questions 1 (2 marks/each):

1. Which cybersecurity property is mainly concerned with keeping information accurate, complete, and protected from unauthorized changes?

A) Confidentiality
B) Integrity
C) Availability
D) Accountability

2. Which statement best distinguishes a worm from a traditional computer virus?

A) A worm only steals passwords, while a virus only encrypts files.
B) A worm requires a host file, while a virus spreads without any user action.
C) A worm can self-propagate across networks, while a virus usually attaches to a host file or program.
D) A worm is always harmless, while a virus is always destructive.

3. A stateful inspection firewall makes filtering decisions mainly by examining packet headers together with:

A) the color of the network cable
B) the current state of the connection/session
C) the user's social media profile
D) the physical location of the server only

4. In the STRIDE threat model, which option best represents Tampering?

A) Pretending to be another user
B) Modifying data, code, or configuration without authorization
C) Denying that an action was performed
D) Making a service unavailable to legitimate users

5. In HTTPS, a digital certificate mainly helps the browser to:

A) increase the website loading speed
B) verify the website identity and support a secure encrypted session
C) remove all malware from the user device
D) replace the need for user authentication

6. Which option is the best example of the privacy principle of data minimization?

A) Collecting all possible user data in case it becomes useful later
B) Keeping personal data forever for future analysis
C) Collecting only the personal data needed for a stated purpose
D) Sharing customer data with all departments by default

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7. Which type of log entry is most useful for detecting repeated password-guessing attempts?

A) Failed login attempts with username, time, and source IP address
B) Daily number of website visitors only
C) The organization's marketing campaign results
D) Printer paper consumption records

8. A laptop containing customer records is lost. Which control most directly reduces the risk of data exposure from the lost device?

A) Increasing screen brightness
B) Using a longer charging cable
C) Full-disk encryption with remote wipe capability
D) Disabling all security updates

9. Which of the following is the clearest example of risk transfer?

A) Removing a risky service completely
B) Buying cyber insurance or contracting a specialist recovery provider
C) Ignoring a critical vulnerability
D) Installing a patch to reduce the likelihood of exploitation

10. In intrusion detection, a false negative occurs when:

A) legitimate activity is incorrectly reported as an attack
B) an attack is blocked before it reaches the network
C) a user forgets a password
D) malicious activity occurs but no alert is raised

PART 2: Short Answer Questions [30 Marks]

P2-Q1 [5 marks]: Compare a stateless packet-filtering firewall and a stateful inspection firewall in terms of how they make filtering decisions and the security benefit of each

P2-Q2 [5 marks]: Why is salted password hashing more secure than storing plain passwords or simple unsalted hashes?

P2-Q3 [5 marks]: Mention two browser-side security practices that reduce the risk of malicious scripts, unsafe downloads, or harmful extensions. Explain how each practice helps.

P2-Q4 [5 marks]: Explain the difference between data minimization and purpose limitation when processing PII. Provide one example for each.

P2-Q5 [5 marks]: Define a false negative in an IDS. Give one possible reason it may happen and one control that can reduce it.

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P2-Q6 [5 marks]: List three important items that should appear in an Acceptable Use Policy (AUP) for company laptops, email, and cloud storage. Then state how audits can verify compliance with the policy.

PART 3: Problem Solving Questions [50 Marks]

P3-Q1 [16 marks]: Given the following access control matrix for an academic services system, construct: (a) the Access Control List (ACL) for each object, and (b) the Capability List for each subject.

Subject/Object	Course_File	Finance_File	Student_DB	Backup_Server
Registrar	Read, Write	-	Read, Write, Own	-
Finance_Officer	Read	Read, Write, Own	Read	-
Lecturer	Read, Write	-	Read	-
System_Admin	Read	Read	Read, Write	Read, Write, Own

P3-Q2 [10 marks]: A university has noticed an increase in malware-related incidents. Several staff computers received suspicious email attachments, some lab machines show unknown processes running at startup, and network logs show connections to suspicious domains. The IT department wants to improve its malware protection strategy based on practical malware protection measures. Answer the following:

- Identify four practical malware protection measures the university should implement, and explain how each measure helps reduce malware infection or limit its impact.
- Recommend two monitoring or response actions the IT team should take when suspicious malware activity is detected.

P3-Q3 [8 marks]: Several university lab computers become very slow. The CPU usage is constantly above 95%, an unknown process starts after login, and network logs show repeated connections to cryptocurrency mining pools. Files are not encrypted. Identify the likely malware category, explain the attack technique, and recommend four mitigation steps.

P3-Q4 [10 marks]: Perform a mini threat-modeling exercise for an online exam submission system. Students upload answer files, teachers download and grade submissions, and administrators manage accounts. Identify four possible threats using STRIDE categories and propose a mitigation for each threat.

P3-Q5 [6 marks]: A typical approach to estimating privacy impact is to look at the two factors that contribute to the impact. Explain these two factors

End of questions

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